The Market: Demand, Supply and Equilibrium

Reference: Intermediate Microeconomics by Hal R. Varian

Indraprastha Institute of Information Technology, Delhi

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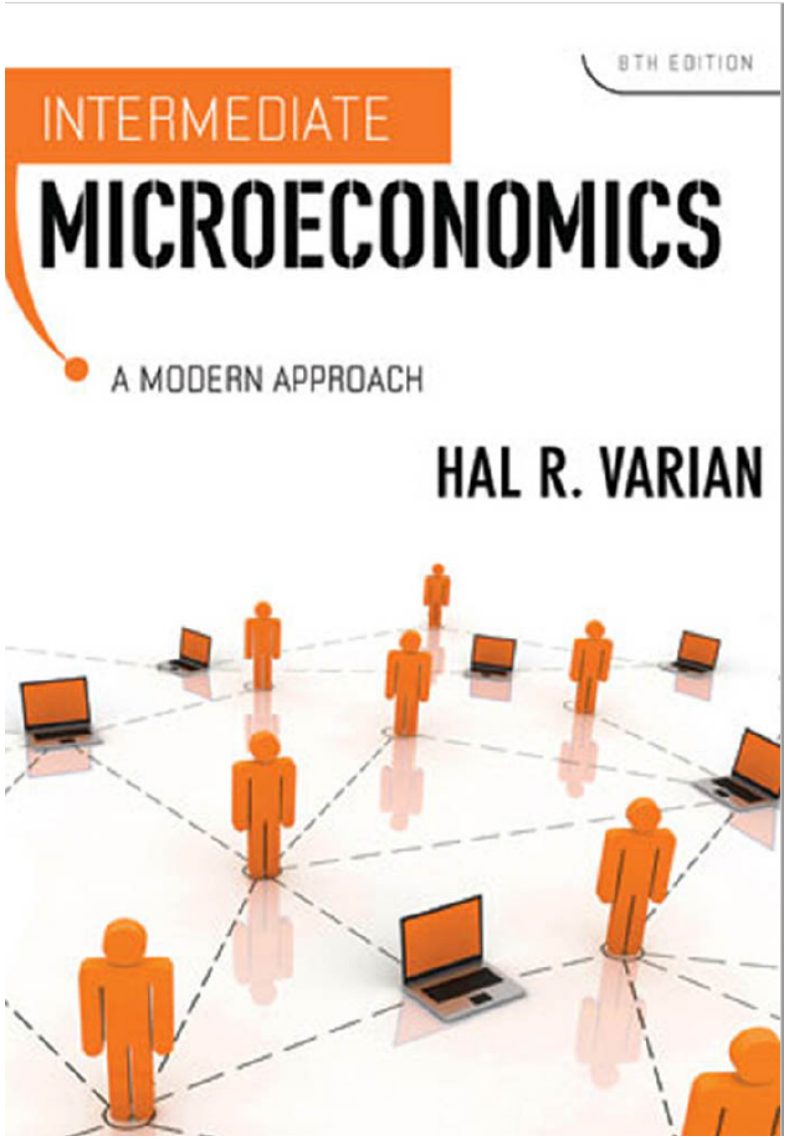
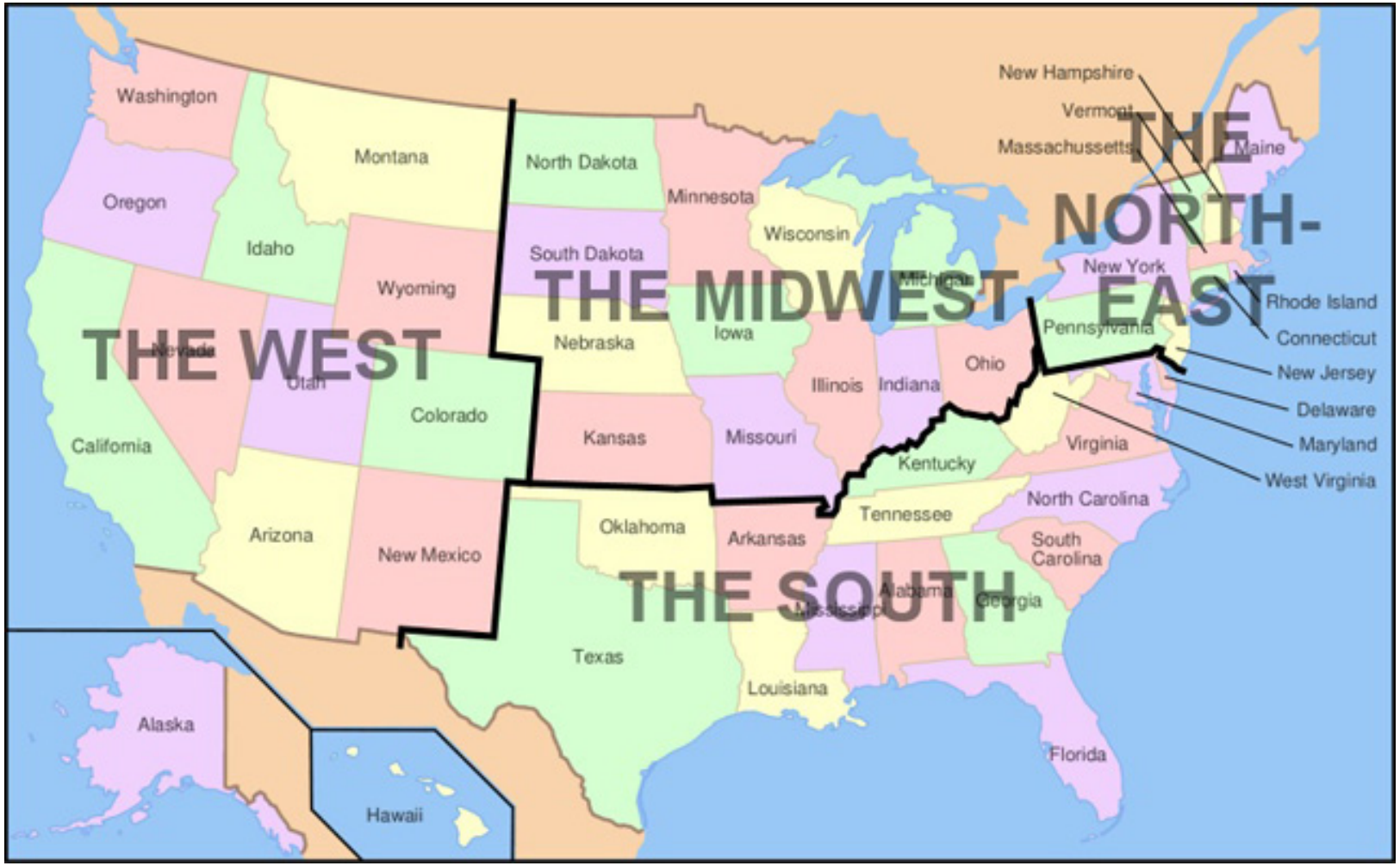

8TH EDITION
INTERMEDIATE
MICROECONOMICS
A MODERN APPROACH
HAL R. VARIAN

Constructing a Model

- Economics: Developing models of social phenomena.
- We will start with an example: market for apartments in a medium sized mid-western college town.
- Types of variables involved:
  - Endogenous variables
  - Exogenous variables

Regions in the USA



**Source: North America Conferences

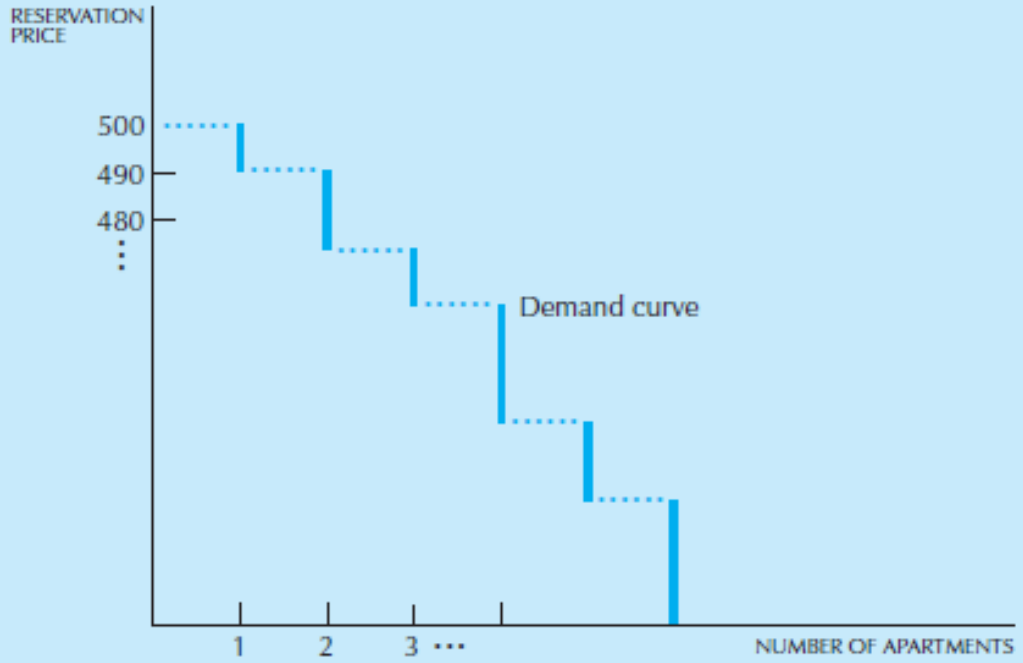
Optimization and Equilibrium

- *The optimization principle:* People try to choose the best patterns of consumption that they can afford
- *The equilibrium principle:* Prices adjust until the amount that people demand of something is equal to the amount that is supplied.

The Demand Curve

- Consider all possible renters of apartments.
- List the highest amount each is willing to pay.
- A person's maximum **willingness to pay (WTP)** is called the person's **reservation price**.
- Let us plot the reservation prices of every individual.

The Demand Curve



The demand curve for apartments. The vertical axis measures the market price and the horizontal axis measures how many apartments will be rented at each price.

Figure 1.1

**Source: Intermediate Microeconomics by Hal R. Varian

The Demand Curve

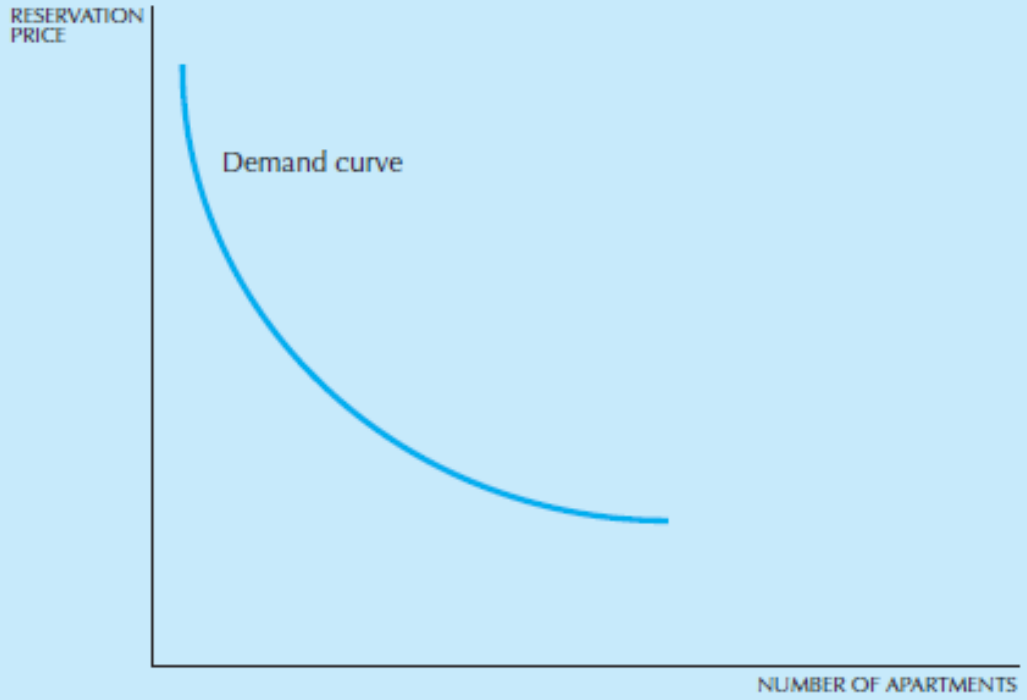


Figure 1.2 **Demand curve for apartments with many demanders.** Because of the large number of demanders, the jumps between prices will be small, and the demand curve will have the conventional smooth shape.

**Source: Intermediate Microeconomics by Hal R. Varian

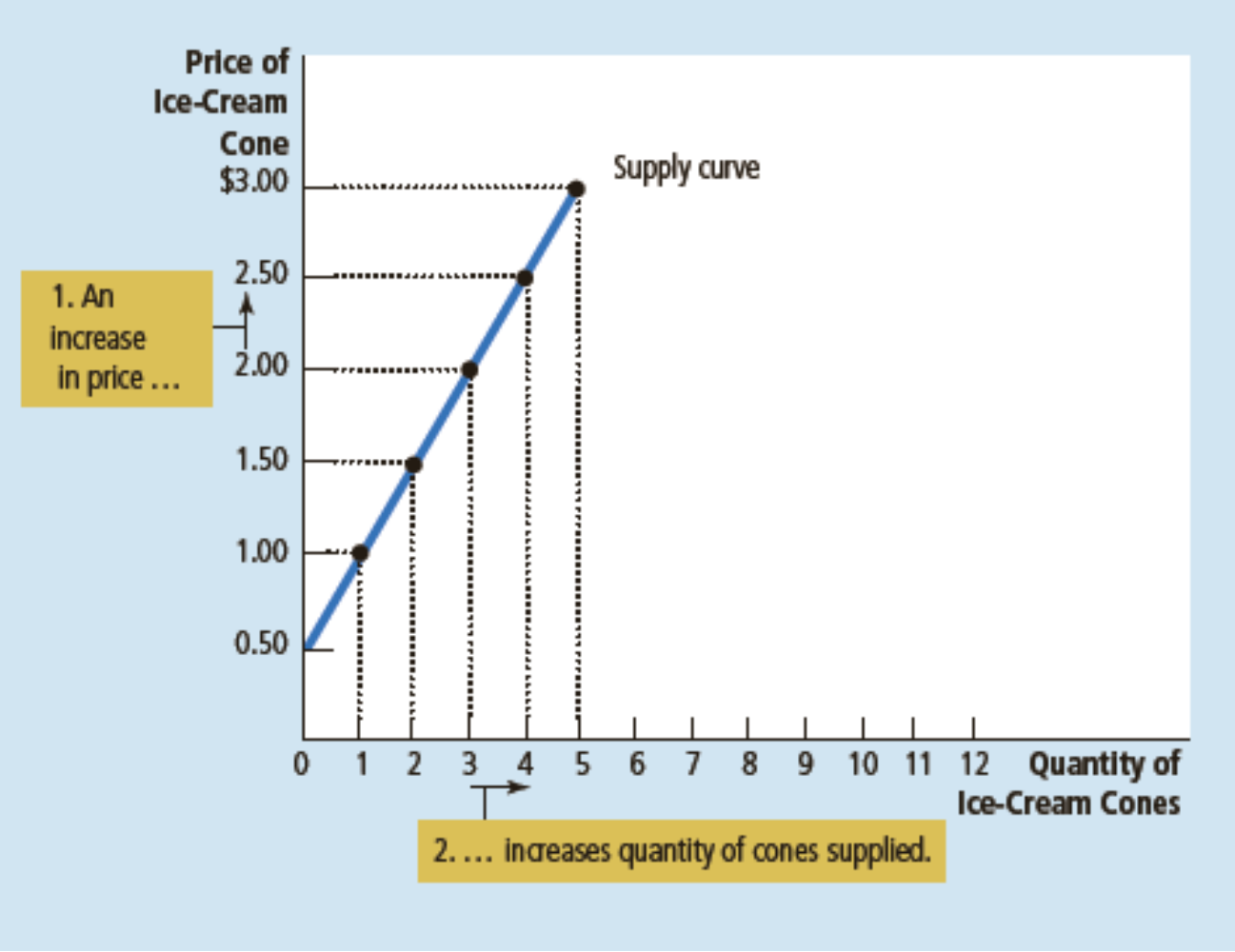
Supply

- Quantity supplied is a sort of mirror of demand, but in a sense, is easier to understand.
- It is the minimum a seller is willing to accept (WTA).
- It's easy to guess how this will reflect in a two dimensional curve.
- Let's understand the idea in detail.

Aggregate and Exogenous Factors

- Supply curves can be aggregated.
- Exogenous variables include
  - Input Prices.
  - Technology (costs).
  - Expectations (A little lesst sraightforward).

The Supple Curve



**Source: Micro by Mankiw (8e)

The Supply Curve: Let's Keep it Siple for now.

- Competitive market.
- Equilibrium price must be the same
- Short run supply is fixed.
- You will see that generally, supply is upward sloping in future chapters. However, in this case, it is fixed.
- This is because, in the short run, the number of apartments cannot change.

The Supply Curve

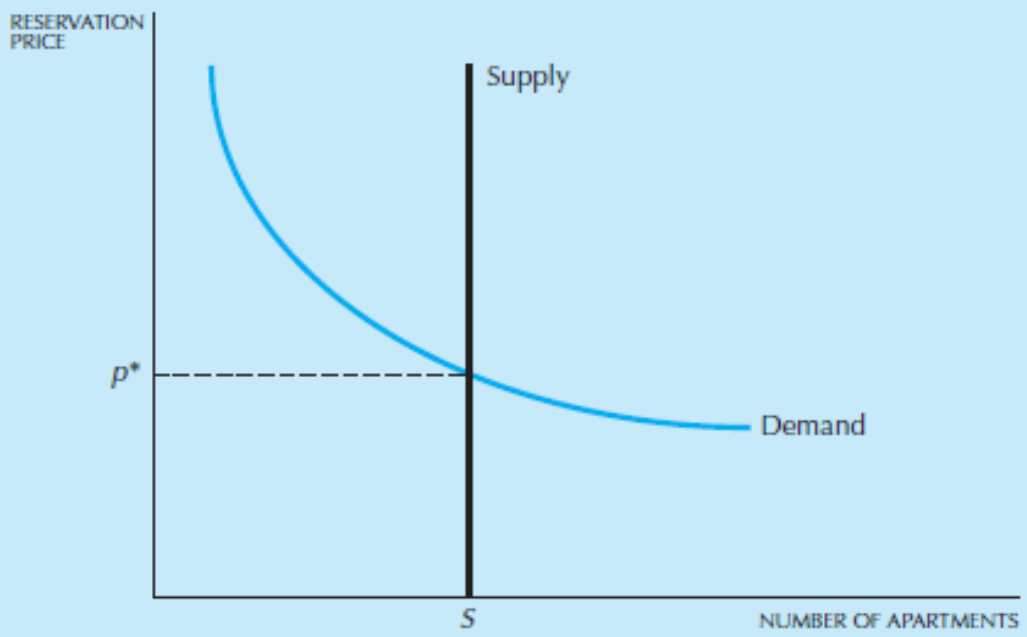


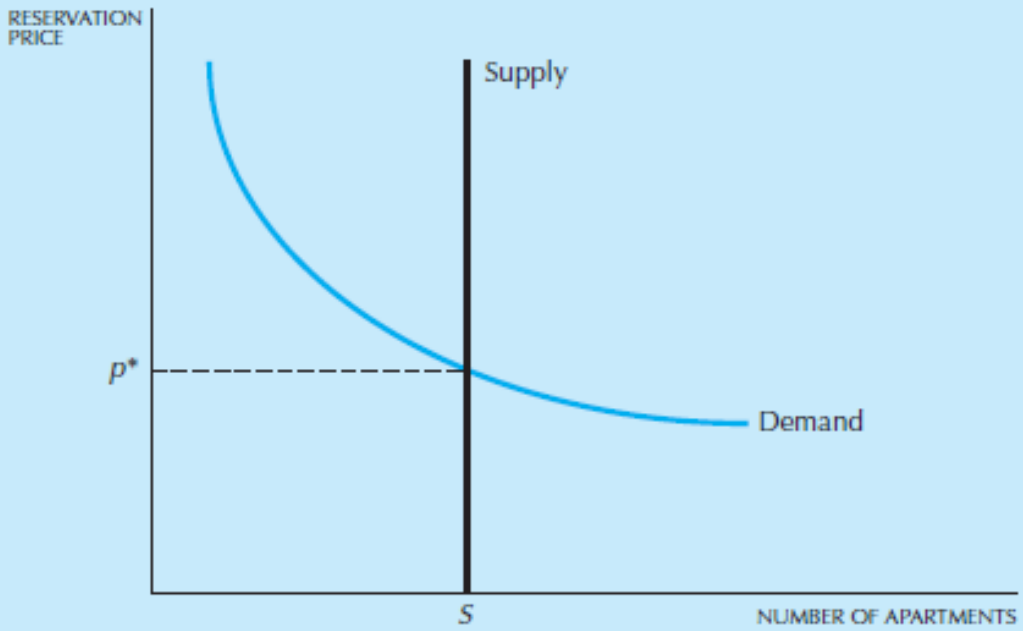
Figure 1.4 **Equilibrium in the apartment market.** The equilibrium price, p^*, is determined by the intersection of the supply and demand curves.

**Source: Intermediate Microeconomics by Hal R. Varian

Market Equilibrium

- Represented by point of intersection between demand and supply.
- Everyone willing to pay the equilibrium price gets an apartment, and the market clears.
- Others choose an alternate option.

Market Equilibrium



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